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Hydro-informer: a deep learning model for accurate water level and flood predictions

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Abstract

This study aims to develop an advanced deep learning model, Hydro-Informer, for accurate water level and flood predictions, emphasizing extreme event forecasting. Utilizing a comprehensive dataset from the Slovak Hydrometeorological Institute SHMI (2008–2020), which includes precipitation, water level, and discharge data, the model was trained using a ladder technique with a custom loss function to enhance focus on extreme values. The architecture integrates Recurrent and Convolutional Neural Networks (RNN, CNN), and Multi-Head Attention layers. Hydro-Informer achieved significant performance, with a Coefficient of Determination (R^2) of 0.88, effectively predicting extreme water levels 12 h in advance in a river environment free from human regulation and structures. The model's strong performance in identifying extreme events highlights its potential for enhancing flood management and disaster preparedness. By integrating with diverse data sources, the model can be used to develop a well-functioning warning system to mitigate flood impacts. This work proposes a novel architecture suitable for locations without water regulation structures.

Keywords Deep learning · Water level prediction · Flood forecasting · Extreme event forecasting · LSTM · GRU · Hydrological attention · Hydrological embedding

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1 Introduction

Floods and droughts are significant hydrological challenges in Slovakia, with substantial impacts on both the environment and society. Recent severe flood events have caused economic losses and posed risks to human lives and infrastructure. Similarly, droughts have disrupted agriculture, water supply, and ecological balance. The increasing frequency and intensity of these events, exacerbated by climate change, underscore the urgent need for accurate prediction models to safeguard the country's resources and population (Brunner et al. 2021; Fendeková et al. 2018).

Current prediction methods for floods and droughts in Slovakia primarily rely on statistical models and traditional hydrological approaches. These methods, while valuable, often fall short in capturing the complex, nonlinear interactions within hydrological systems, especially under extreme conditions. Their limitations in accuracy and reliability highlight the pressing need for advanced predictive tools that can provide timely and precise forecasts (Krzysztofowicz 1999; Golding 2011; Helsel et al. 2018).

Despite advancements in hydrological modeling, there remains a significant gap in applying state-of-the-art deep learning techniques to flood and drought prediction in Slovakia. Most existing studies focus on conventional methods, with limited exploration of how modern computational approaches can enhance prediction accuracy and reliability. This gap represents a missed opportunity to leverage the powerful capabilities of deep learning models, which have shown promise in handling complex and large datasets across various fields (Tripathy and Mishra 2024; Sit et al. 2020). Our research aims to bridge this gap by developing and applying advanced deep-learning architectures tailored to the specific hydrological conditions in Slovakia.

Deep learning models, particularly Long Short-Term Memory Networks (LSTM), have emerged as some of the most accurate tools for rainfall-runoff predictions. However, concerns persist regarding their reliability in predicting extreme events. Frame et al. (2022) tested LSTM networks and a mass-conserving LSTM variant, finding that these models remained relatively accurate in extreme event predictions compared to traditional models like the Sacramento Model and the US National Water Model. Hybrid models that integrate different deep learning architectures, such as CNN-LSTM models, have demonstrated significant improvements in capturing complex temporal and spatial dependencies in hydrological data. For instance, Li et al. (2022) developed a CNN-LSTM model for river flow prediction, which outperformed models using either CNN or LSTM alone. This hybridization effectively identifies spatial and temporal features in precipitation data, essential for accurate forecasting.

Despite these advancements, deep learning models still face challenges, particularly in handling longer forecast horizons, data quality issues, and the inherent complexity of extreme weather events. For example, LSTM models tend to show diminished accuracy for forecasts extending beyond 12 h, especially during the rainy season (Zhang et al. 2022). Similarly, while CNN-LSTM models improve spatial and temporal dependency capture, their performance is often constrained by input feature quality, necessitating further refinement for more reliable predictions. Additionally, deep learning models struggle with data imbalance, model interpretability, and computational resource demands, limiting their generalizability and real-world applicability (Dong et al. 2020). Addressing these challenges through the integration of advanced architectures, improved data handling, and model optimization is crucial for enhancing the accuracy and reliability of flood and drought predictions (Luppichini et al. 2023).

The primary objective of this research is to develop and implement a novel hybrid deep learning model that integrates CNN, LSTM, and attention-based architectures with a gating mechanism. This model is specifically designed to predict extreme hydrological events, focusing on regions with no water regulations, where data scarcity and poor quality have traditionally limited predictive accuracy. Integrating these advanced models with existing hydrological data is expected to significantly improve decision-making in water resource management and disaster mitigation, thereby enhancing disaster preparedness and risk management (Tripathy and Mishra 2024; Luppichini et al. 2022; Mosavi et al. 2018). Our research will explore and compare the effectiveness of various deep learning architectures, including LSTM, CNN, Transformer-based architectures, and Fully Connected Neural Networks (FCNN), in forecasting hydrological events.

Enhancing the accuracy and reliability of flood and drought predictions is crucial for effective decision-making and mitigation strategies. This research aims to provide essential insights for policymakers and stakeholders, contributing to a more resilient Slovakia, better equipped to manage and mitigate the impacts of extreme hydrological events (Almikaeel et al. 2022) and climate change (Repel et al. 2021; Sleziak et al. 2021).

2 Methods

2.1 Data collection and preprocessing

The dataset utilized for this study was obtained from the SHMI, encompassing a comprehensive range of hourly measurements, including water levels (measured in centimeters), discharge rates (measured in cubic meters per second), and precipitation levels (measured in millimeters) of the Topl'a River, a prominent watercourse in eastern Slovakia, which serves as the right tributary of the Ondava River. The hydrological situation in Slovakia reflects its rich geographical diversity. River discharges in Slovakia peak during late spring and early summer due to the combined influence of spring snowmelt and rainfall, while low flows are typically observed in late summer and autumn. Flood events, particularly flash floods during summer thunderstorms, are common during these high discharge periods. The Topl'a River spans a catchment area of 1544 km^2 and a length of 129.8 km. It experiences a temperate climate with annual rainfall ranging from 600 to 1000 mm, depending on elevation. Significant flooding events have occurred over the years, including in May 1987 (Q = 235 m^3/s, overcome in 2010), June 2006 (Q = 53 m^3/s, a 2-year flood), July 2008 (Q = 169 m^3/s, a 20-year flood), and June 2010 (Q = 350 m^3/s, a 100-year flood), causing severe erosion and disruption of the river's natural meandering. This study focuses on the Topl'a River's hydrological patterns, aiming to improve flood prediction and management in this region (Frandofer and Lehotský 2014; Fendeková et al. 2018). Figure 1 shows the location of the Topl'a River within Slovak territory, as well as the gauging station and its location along the river.

The dataset spans from 2008 to 2020, offering rich temporal resolution essential for accurate predictive models. However, numerous gaps and inaccuracies presented significant preprocessing challenges. A primary issue was erroneous data points; for example, water level readings sometimes spiked to 800 cm without corresponding precipitation or hydrological events. These anomalies indicated measurement errors or data entry mistakes, as such extreme levels were not supported by historical records.

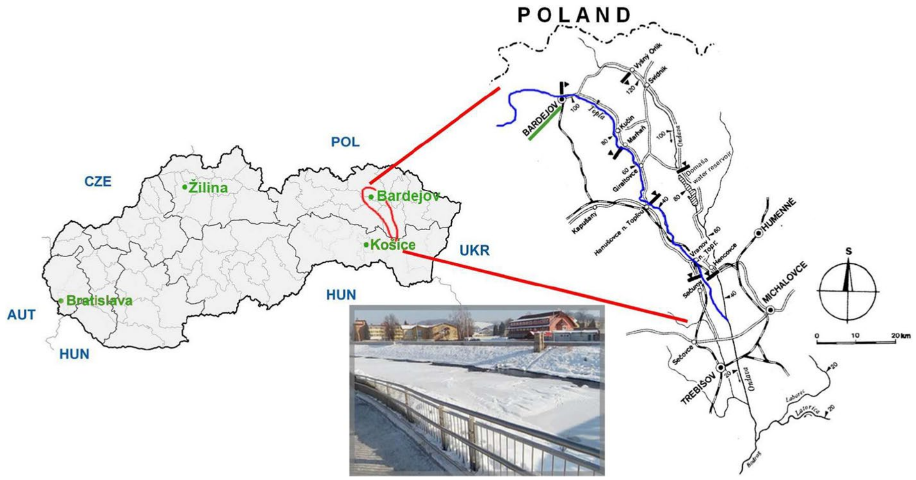

Fig. 1 Location of the Topl'a River in Slovakia, including the gauging station used for data collection

To address these issues, we employed a rigorous data preprocessing strategy involving several critical steps:

- *Removal of Inaccurate Values* All identified inaccurate values were systematically removed, including outliers such as extreme water level spikes and any data points that did not align with known hydrological patterns or lacked supporting meteorological evidence.
- *Handling Missing Data* Given the prevalence of missing values, we removed the corresponding records to maintain dataset integrity. This step was essential to prevent biases or inaccuracies from attempting to impute or interpolate missing data in a highly variable dataset with extreme values.
- *Homogeneity Check* After removing inaccurate and missing data, we conducted a homogeneity check to ensure the remaining data followed expected statistical distributions, particularly focusing on a log-normal distribution common in hydrological data. This step was crucial for ensuring the dataset's suitability for modeling, as homogeneous data improves the reliability and accuracy of predictive models.
- *Final Validation* The homogeneity check also served as a final validation to confirm that the data-cleaning process had not introduced new inconsistencies. Ensuring the dataset adhered to expected statistical properties increased our confidence in the robustness of subsequent modeling efforts.

The final dataset included hourly records for precipitation, discharge, and water level measurements from 2008 to the end of 2015. However, a significant gap existed from early 2016 to late 2017 due to multiple data inconsistencies and errors. During this period, extreme increases in one variable often occurred without corresponding changes in others, and some data points were missing. From 2018 to 2020, the dataset resumed with complete and accurate hourly records for a full 2-year period. This cleaned and processed dataset forms a robust foundation for developing and validating predictive models to enhance water level and flood forecasting accuracy.

To transform the preprocessed data into model-ready formats, we developed a function that converts DataFrame columns into NumPy arrays (Harris et al. 2020) for efficient processing and handles the creation of input and output sequences with a specific overlap strategy. This function also introduces a small amount of random noise to the precipitation array to account for variability. The processed data was divided into training and testing sets to ensure robust model evaluation. Training data included all years except 2008 and 2014, which were reserved for testing. This method ensured the model encountered diverse conditions and significant water level events. By training on a wide range of conditions and testing on notable water level events, the approach provided a thorough evaluation of the model's performance.

The data was divided to ensure the testing set included a variety of extreme and normal water levels, allowing for a comprehensive assessment of the model's performance on unseen data. This approach is crucial for verifying the model's capabilities and reliability in predicting both regular and extreme hydrological events. The training set encompassed a broad spectrum of low, moderate, and extreme values, ensuring exposure to various conditions during training. This exposure helps the model learn the underlying patterns, enhancing its ability to generalize and apply learned patterns to the testing data effectively.

The input data shapes for the training set consisted of 4923 samples, each with a sequence length of 36 h and three variables (precipitation, discharge, and water level) with the following shape (4923, 36, 3). The output data shapes were 4923 samples, each with a sequence length of 12 h and one variable (predicted water level) with the corresponding shape (4923, 12, 1). The testing data followed a similar structure but with 1456 samples each. These dimensions reflect the sequence lengths and the number of features used for modeling.

Data collection and preprocessing were vital in transforming a raw and noisy dataset into a clean and reliable foundation for developing advanced predictive models. Meticulous removal of inaccuracies and gaps, coupled with rigorous statistical validation, ensured the dataset was accurate and representative of underlying hydrological processes. This prepared the ground for the subsequent application of deep learning techniques, enhancing the accuracy and reliability of flood and water level predictions.

3 Model development

3.1 Hydro-informer layers

- *Embedding layers* transform raw input data, such as precipitation, discharge, and water levels, into a higher-dimensional space. This transformation helps the model capture complex relationships, making it easier to forecast hydrological events.
- *Concatenation* combines different types of input data (e.g., precipitation, discharge, water levels) into a single dataset. Scaling adjusts these combined data points so they can be processed consistently by the model. This step ensures balanced input, which is essential for accurate forecasting.
- *Conv1D layers* apply filters to detect temporal patterns in the data, such as daily or hourly changes in water levels. These filters help the model identify important trends and features necessary for precise hydrological predictions.
- *Spatial dropout* is a regularization technique used to improve the model's ability to generalize from the training data to real-world scenarios. By randomly ignoring parts

of the input data during training, it prevents the model from relying too heavily on any single data source, creating a robust model that can handle varied and unexpected inputs.

- *Gated Recurrent Units GRU and LSTM* layers are designed to understand sequences of data over time. In hydrological forecasting, they help the model remember past water levels and precipitation patterns and use this information to predict future conditions. Their bidirectional nature allows them to consider both past and future data points, enhancing the accuracy of forecasts.
- *Multi-head attention* allows the model to focus on multiple aspects of the input data simultaneously. For hydrological forecasting, this means the model can consider various factors like precipitation intensity and river flow at the same time, improving the overall forecast by understanding how different factors interact and influence each other.
- *Gating mechanisms* help the model decide which information is most important at any given time. By filtering out less relevant data, the model can concentrate on the most critical factors affecting water levels and flood risks, leading to more accurate predictions.
- *Residual connections* allow the model to pass information directly from one layer to the next without losing valuable details, making training more efficient. Layer normalization standardizes the inputs to each layer, ensuring that the model learns effectively from the data. Together, these techniques maintain stability and improve the accuracy of forecasts.
- *Time-distributed dense layers* apply processing to each time step in the data sequence individually. This is particularly useful for making predictions at each hour or day based on historical data. By handling each time step separately, the model can provide detailed and precise forecasts.
- *Reshape and flatten* operations prepare the data for final processing. Reshaping adjusts the data into the right format for the next layer, while flattening simplifies it into a single list of values. These steps ensure that the data flows smoothly through the model, enabling it to make accurate hydrological predictions.

Each of these components plays a crucial role in building a robust model for hydrological forecasting, helping to predict events like floods and water level changes with greater accuracy and reliability (Chollet et al. 2015).

3.2 Hydro-informer blocks

The Hydro-Informer is a sophisticated deep learning framework designed to predict water levels, focusing on accurately forecasting extreme events. The data includes hourly measurements of water levels, discharge rates, and precipitation levels, segmented into sequences of 36-hour inputs to predict the subsequent 12 h. The model architecture integrates LSTM and GRU to manage sequential dependencies in the time series data. One-dimensional Convolutional Neural Networks (Conv1D) are employed to extract local patterns from these sequences. Multi-head attention mechanisms are incorporated to enhance the model's ability to focus on relevant features, thereby improving prediction accuracy. During training, a custom loss function emphasizes the accurate prediction of extreme values, ensuring the model prioritizes critical flood-indicating data points. This comprehensive processing approach enables the model to deliver precise and reliable water level forecasts, vital for effective flood management and early warning systems (Fig. 2).

3.2.1 Feature extractor/hydro encoder

The Feature Extractor/Hydro Encoder block is crucial for extracting meaningful features from historical data inputs, capturing both local and global dependencies to produce refined encoded features for further processing.

- *Conv1D Layer* Applies filters to extract local features along the time axis, capturing short-term dependencies and patterns essential for precise hydrological predictions.
- *Spatial Dropout* Prevents overfitting by randomly dropping portions of the convolutional output during training, ensuring the model does not rely too heavily on any single feature.
- *Bidirectional GRU + LSTM* Combines GRUs and LSTM to capture dependencies from both past and future time steps, providing a comprehensive understanding of the sequence context.
- *Multi-Head Attention* Focuses on different parts of the input sequence simultaneously, computing attention scores for multiple heads, concatenating the results, and projecting them back into the original feature space to learn complex dependencies and relationships within the data.
- *Gating Mechanism* Uses a sigmoid activation to compute a gate value, selectively focusing on important features in the input sequence and filtering out less relevant information.
- *Residual Connections + Layer Normalization* Passes information directly from one layer to the next without losing valuable details, making training more efficient. Layer normalization stabilizes training by normalizing inputs to each layer.
- *Additional LSTM + Conv1D Layers* Further refine features extracted from previous layers, capturing more intricate patterns. LSTM layers handle temporal dependencies, while Conv1D layers extract local features.
- *Encoded Features* Apply dense layers to each time step independently, adjusting the output shape to match the target sequence length, ensuring consistent feature representation across time steps.

3.2.2 Hydro-decoder

- *Multi-Head Attention* Integrates encoded features from both data sources, focusing on different sequence parts simultaneously. This mechanism computes attention scores, combines the results, and projects them back, enhancing the model's ability to capture complex dependencies.
- *Add Attention to Encoded Features* Merges processed outputs using attention mechanisms, ensuring the model leverages the most relevant information from both historical and forecasted data sources.
- *Time-Distributed Dense Layers* Produces final predictions by applying dense layers to each time step independently, ensuring consistent and accurate predictions for each time step in the sequence.

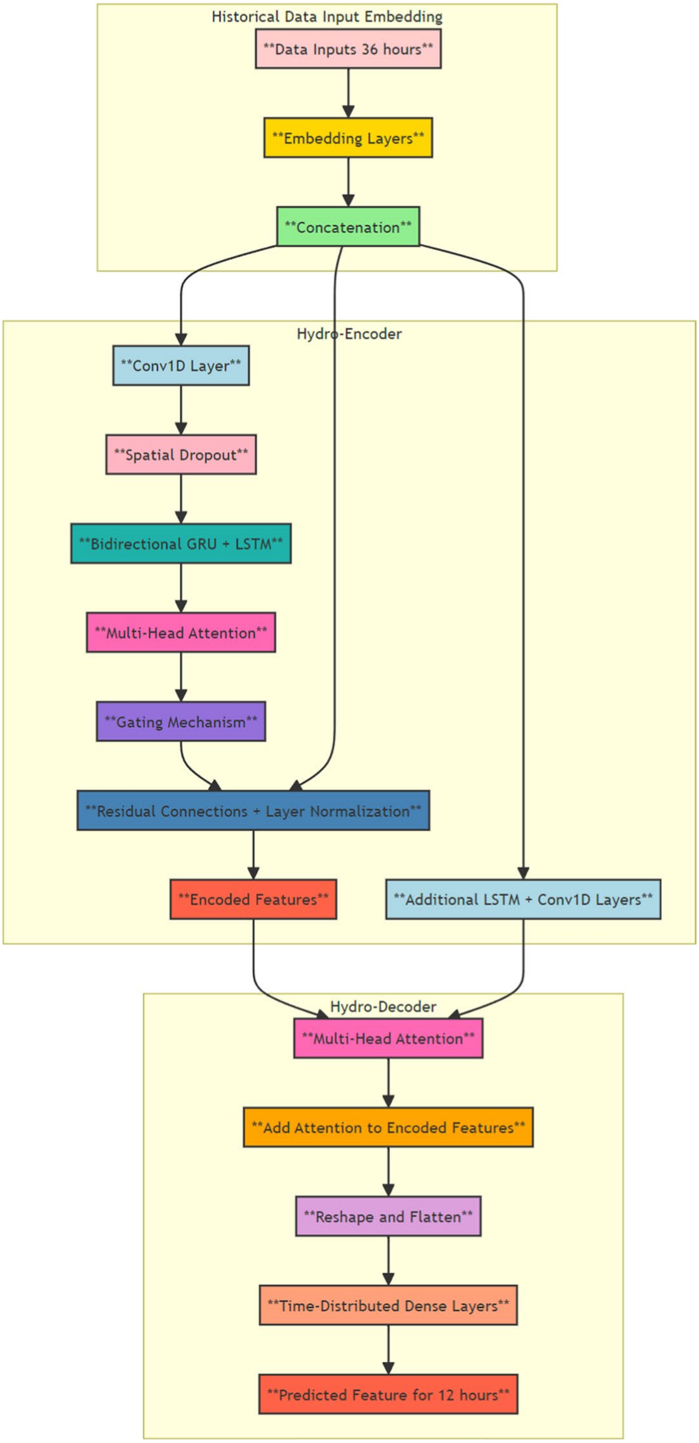

Fig. 2 Hydro-informer: architecture and blocks

- *Output Layer* Provides the final predicted features for the next 12 h, representing the model's forecast, incorporating all refined features to generate accurate and reliable predictions.

The Hydro-Decoder block processes encoded features from the encoder and row-embedded data with simple processing, leveraging the most relevant information from both sources. The Multi-Head Attention mechanism focuses on different parts of the combined sequences simultaneously, enhancing the model's ability to capture complex dependencies and relationships. The resulting predictions are highly accurate, providing valuable insights into future trends and conditions, making this model an effective tool for time series forecasting in hydrology and other fields.

3.3 Hyperparameter tuning and optimization techniques

The custom loss function developed for this study focuses on accurately predicting extreme water levels, essential for effective flood warning and mitigation. Traditional loss functions, like (MSE), treat all errors equally, which can lead to suboptimal performance when predicting extreme values is more critical. To address this, we created a custom loss function that assigns higher penalties to errors associated with extreme values. This ensures the model prioritizes accurately predicting extreme values, which are crucial for flood risk management.

The extreme threshold is set at 200 cm. Any predicted value above this threshold that underestimates the true water level incurs a significant penalty during training. This emphasis trains the model to be sensitive to potential flood events, crucial for timely and effective response. Additionally, a 20 cm threshold was set to avoid significant overestimation or underestimation within this range. Deviations within 20 cm result in the loss being multiplied by a factor of 10, encouraging the model to adjust its parameters and improve prediction accuracy.

3.4 Training process

Initially, data preprocessing was conducted to clean and normalize input values, ensuring dataset consistency. The function generated the input, output, and auxiliary arrays. Training data shapes were (4923, 36, 3) for inputs, and (4923, 12, 1) for outputs. The testing data had 1456 samples each. This preprocessing was crucial for effective training and evaluation.

Various deep learning models, including LSTM, GRU, Conv1D, and Multi-Head Attention layers, were initialized with suitable hyperparameters. Combining RNNs like LSTM and GRU with Conv1D layers effectively captured and analyzed temporal patterns. Multi-Head Attention mechanisms enhanced the model's focus on relevant features, even with lower-quality data.

A custom loss function, was implemented to prioritize the accurate prediction of extreme water levels. This function assigned higher penalties to errors for extreme values (water levels above 200 cm) and underestimations within 20 cm. This strategy ensured the model focused on predicting extreme events accurately while maintaining precision within a critical range.

The training process employed a ladder technique, where the model was trained in four phases with varying thresholds and extreme value limits to progressively enhance its predictive capabilities:

- *Phase 1* The model was trained with `threshold=30` and `extreme_threshold=300` for 20 epochs. This phase focused on enabling the model to accurately predict smaller water levels.
- *Phase 2* Using the weights from Phase 1, the model was further trained with `threshold=25` and `extreme_threshold=250` for 20 epochs. This phase aimed to improve the model's ability to predict medium to large water levels while maintaining accuracy for smaller values.
- *Phase 3* Continuing from Phase 2, the model was trained with `threshold=15` and `extreme_threshold=200` for 20 epochs. This phase enhanced the model's performance in predicting large water levels above 200 cm, with a margin of error within 15 cm for smaller and medium values.
- *Phase 4* Finally, the model was trained with `threshold=10` and `extreme_threshold=200` for 20 epochs using the weights from Phase 3. This phase refined the model's capability to predict large water levels above 200 cm accurately, within a 10 cm error margin for all values, ensuring the model could generalize well and predict peaks on time or even before they occurred.

Hyperparameter tuning involved experimenting with various configurations, such as the number of layers, units per layer, learning rate, and batch size. The model was trained with a batch size of 16 and a validation split of 0.1 for 20 epochs. Early stopping and model checkpointing techniques were employed to prevent overfitting and to save the best-performing model based on validation loss.

4 Results

4.1 Model performance

The performance of the Hydro-Informer model was evaluated using several key statistical metrics to ensure its accuracy and reliability in predicting water levels and flood risks. Table 1 summarizes the performance metrics of the model.

Table 1 Model performance metrics

Metric	Value (units)
Mean squared error (MSE)	18.0704 cm^2
Root mean squared error (RMSE)	4.2509 cm
Mean absolute error (MAE)	1.8226 cm
Mean absolute percentage error (MAPE)	1.11%
Coefficient of determination (R^2)	0.8785
Mean squared logarithmic error (MSLE)	0.0005
Root mean squared logarithmic error (RMSLE)	0.0227
Symmetric mean absolute percentage error (sMAPE)	13.30%

Metrics without specified units are dimensionless

- *MSE* The MSE was 18.0704. It measures the average squared difference between actual and predicted values. A lower MSE indicates better model accuracy, showing a relatively low error in predictions.
- *RMSE* The RMSE was 4.2509. It provides the error in the original units (centimeters), highlighting larger errors. An RMSE of 4.2509 cm suggests the model's predictions are, on average, within 4.25 cm of the actual water levels.
- *MAE* The MAE was 1.8226. It calculates the average magnitude of errors without considering their direction. An MAE of 1.8226 cm indicates high accuracy, with the average error being just under 2 cm.
- *MAPE* The MAPE was 1.11%. It expresses the prediction error as a percentage, showing that, on average, the model's predictions are within 1.11% of the actual water levels, reflecting high predictive accuracy.
- R^2 The R^2 was 0.8785. It represents the proportion of variance in water levels that the model can explain. An R^2 of 0.8785 indicates a strong fit, capturing most of the variance.
- *MSLE* The MSLE was 0.0005. It is useful for predicting small values accurately, penalizing underestimations more. The low MSLE value suggests the model captures log-transformed variations in water levels well.
- *RMSLE* The RMSLE was 0.0227. It provides a logarithmic perspective on errors, useful for comparing relative errors. The low RMSLE further confirms the model's accuracy in predicting water levels on a logarithmic scale.
- *sMAPE* The sMAPE was 13.30%. It normalizes the error, making it symmetric and bound between 0 and 200%. A sMAPE of 13.30% indicates a reasonable balance between overestimation and underestimation errors.

Despite strong metrics, the model has some limitations. The high sMAPE indicates variability in accuracy across different water levels, suggesting performance issues at the extremes. While the custom loss function prioritizes extreme values, it may bias normal-level predictions. Data quality and completeness significantly impact performance, as gaps or inaccuracies reduce reliability. Additionally, the model's dependence on finely tuned hyperparameters means that dataset changes could require re-optimization to maintain performance.

Figure 3 illustrates a scatter plot comparing predicted water levels to actual water levels. The red dashed line represents the perfect prediction line, where predicted values match actual values exactly. The shaded areas below this line indicate regions of underestimation, with the yellow and orange zones representing the severity of potential risks associated with prediction errors.

This figure demonstrates that most predictions are close to actual values, although there are instances of underestimation, particularly at higher water levels. The model's primary purpose is to predict extreme values or outliers, where underestimation poses critical risks. While overestimations can often be addressed through calibration models, underestimations-especially in flood forecasting-pose significant risks. Two underestimation zones have been classified in this study to assess the risk associated with the model predictions:

(1) *Critical Underestimation (light orange)* This zone indicates predictions that fall below actual water levels by up to 70 cm. Such underestimation is concerning, especially when actual water levels exceed 300 cm, as it may lead to inadequate flood prepared-

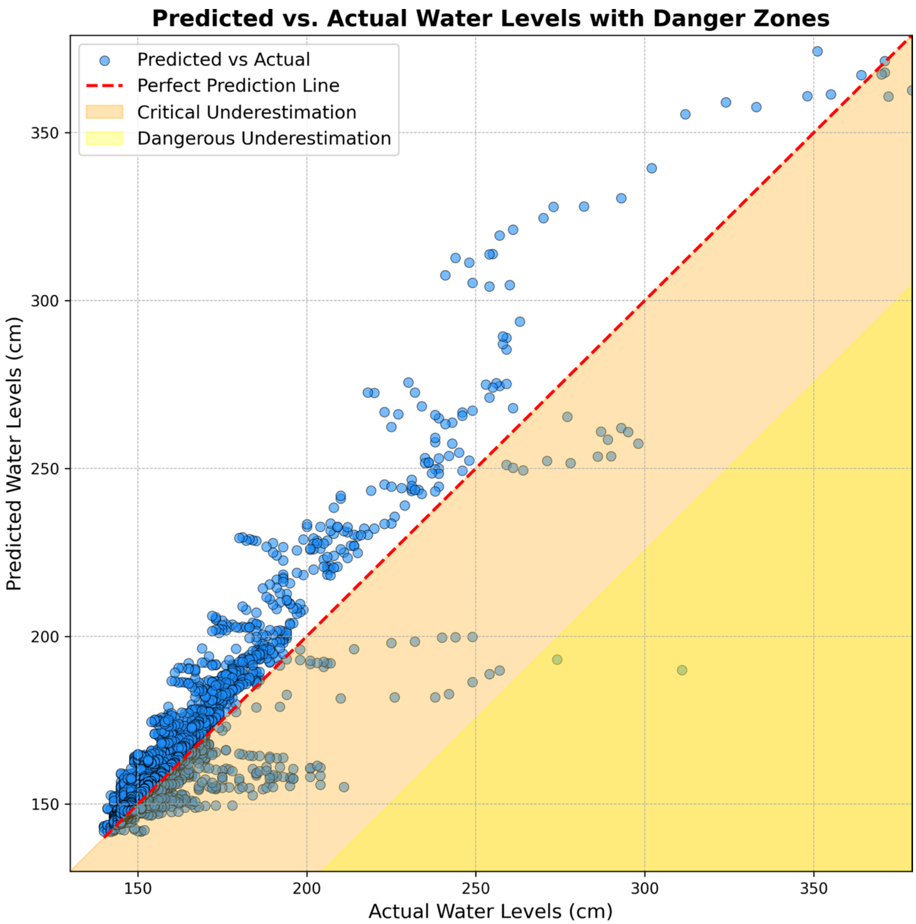

Fig. 3 Predicted versus actual water levels with danger zones

ness. Nevertheless, the model demonstrates robustness in predicting within the 250–300 cm range, with limited underestimation.

(2) *Dangerous Underestimation (yellow)* This zone represents predictions that are more than 70 cm below actual water levels, indicating severe underestimation. Such significant discrepancies could result in under-preparation for flood events, posing substantial risks to public safety and infrastructure.

These zones are defined based on the potential consequences of underestimating water levels, with thresholds tailored to local conditions and historical data. While the model generally performs well, with most predictions close to the perfect prediction line, some instances fall into these underestimation zones, particularly at higher water levels. This highlights the need for ongoing model refinement, particularly for predicting extreme events, to minimize critical and dangerous underestimations.

The quality of the data used in the training is a significant factor in these results. The lack of diverse data types also affects the accuracy and predictability of the model. For extreme events with water levels of 300 cm or more, the model was able to identify all extreme values except for two non-consecutive data points. These discrepancies are more likely to be originated from poorly recorded values or data entry errors. This suggests that while the model performs well overall, there is still room for improvement, particularly in handling extreme values and addressing data quality issues.

Figure 4 shows the performance of the predictive model, comparing actual water levels (solid blue line) to predicted values (dashed orange line) over time. A light blue shaded area represents the confidence interval, derived from the standard deviation of the predictions, indicating the range within which true water levels are expected to lie.

The confidence interval is scaled using a factor of 0.5 for optimal visibility and accuracy, ensuring it is neither too narrow nor too wide. This interval captures the general trend and variability of actual water levels, reflecting the model's precision and inherent uncertainty in its predictions. While most actual values fall within this interval, deviations occur, especially during extreme events.

Figure 4 highlights the model's ability to track actual water levels closely, with most predictions falling within the confidence interval. However, discrepancies during extreme water levels indicate areas for further refinement.

Overall, the figure suggests that the model is generally reliable for normal conditions but may require additional calibration for extreme events. This is crucial for practical applications like flood forecasting and water resource management, where precise predictions are essential for timely and effective decision-making.

4.2 Analysis of extreme values

In the context of flood management and early warning systems, accurately predicting extreme water levels is critical. This section evaluates the performance of the model in predicting the three largest and most dangerous peaks in the testing dataset, focusing on both the magnitude and timing of these events. These factors are essential for effective flood management and timely warnings.

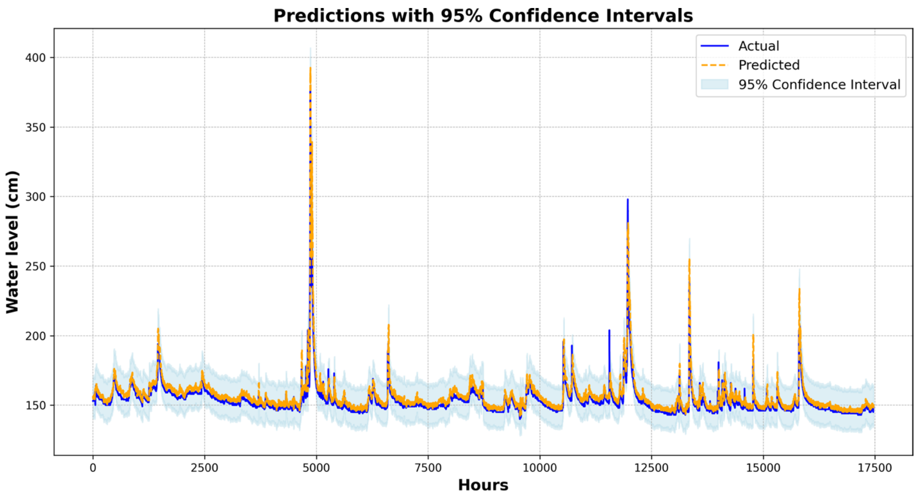

Fig. 4 Predictions with adjusted confidence intervals

The Hydro-Informer produces predictions for the next 12 h based on the previous 36 h of data. The data was divided into inputs and outputs with a 12-hour shift window, meaning the information was updated every 12 h, and the results were produced for the subsequent 12 h. To evaluate the performance of the model, tests were conducted using hourly data from the years 2008 and 2014, which were included in the testing dataset. These years were specifically chosen due to the occurrence of significant water level peaks, providing a rigorous testing ground for the model's predictive capabilities in critical situations.

As illustrated in Fig. 5, the study focuses on the top three peaks, which are the only significant peaks observed in the testing data. These peaks represent the most challenging scenarios for the model, highlighting its ability to accurately predict extreme hydrological events and assess its performance under such conditions.

4.2.1 Extreme analysis—peak 1

In the first case, illustrated in Fig. 6, the actual peak water level was approximately 379 cm. The model predicted a peak of around 374.19 cm occurring 2 h before the actual peak. The difference in peak value is 4.79 cm, showing a relatively minor underestimation. The timing difference (ΔTime) of 2 h again demonstrates the model's capability to predict peaks slightly ahead of the actual event. This capability is crucial for issuing timely alerts, even though the slight magnitude difference requires further calibration for precision.

4.2.2 Extreme analysis—peak 2

The second peak, depicted in Fig. 7, reached an actual level of approximately 298 cm, with the predicted peak being around 274.92 cm. The model predicted this peak 1 h before the actual peak occurred, resulting in a ΔPeak Value of 23.08 cm. This larger difference indicates a significant underestimation, which could be due to the limitations in the dataset quality or the model's ability to handle extreme values accurately. Nevertheless, the

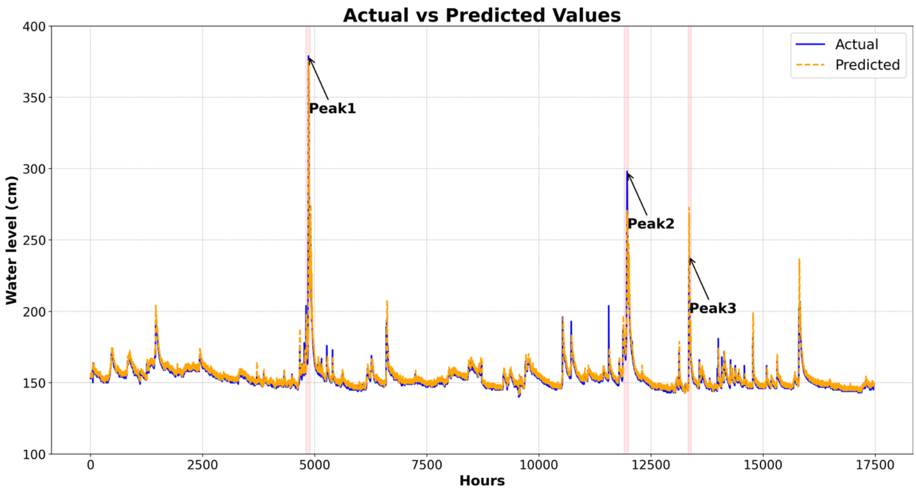

Fig. 5 Comparison of actual and predicted water levels over time, highlighting the three most significant peaks

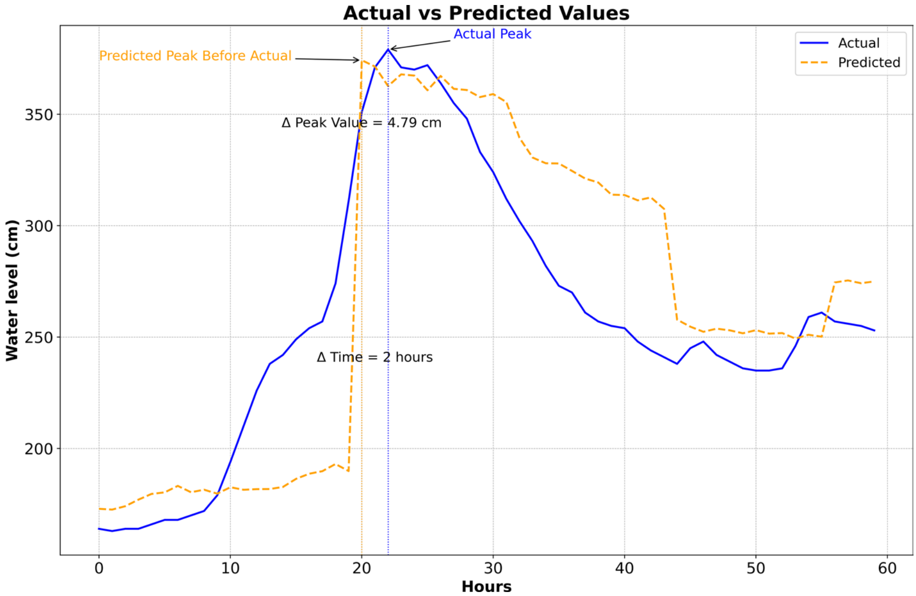

Fig. 6 Peak analysis case 1: actual versus predicted water levels

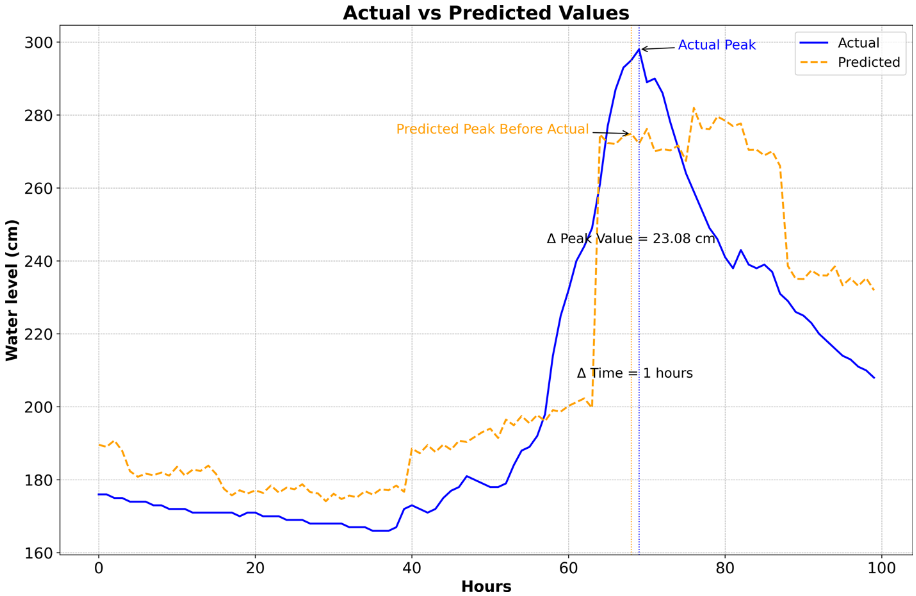

Fig. 7 Peak analysis case 2: actual versus predicted water levels

early prediction (ΔTime = 1 hour) provides a short but critical lead time for emergency responses.

4.2.3 Extreme analysis—peak 3

The third peak analyzed is shown in Fig. 8. The actual peak water level reached approximately 239 cm, while the predicted peak before the actual peak occurred was around 246.66 cm. The difference in peak value (ΔPeak Value) is − 7.66 cm, indicating that the model slightly overestimated the peak. However, the predicted peak occurred 5 h earlier than the actual peak (ΔTime = 5 h). This early prediction is advantageous for early warning systems, as it provides a lead time for flood preparation and response, although the magnitude overestimation needs addressing for more precise warnings.

These extreme analyses demonstrate the model's capability to predict peak water levels before they occur, providing valuable lead time for flood warnings and management. The early prediction of peaks, even with some underestimation, is a significant advantage for early warning systems, allowing for timely preparations and potentially mitigating flood impacts.

Every value predicted on the graph is predicted 12 h in advance, reflecting the model's capability of detecting outliers 12 h in advance. In some cases, within the range of the 12-h predictions, the model was able to detect the peak even earlier, suggesting that the model has the potential to identify rising trends and anticipate extreme values before they fully develop. This early detection is crucial for proactive flood management and emergency response.

However, the analysis also highlights limitations in the model's accuracy, particularly in predicting the exact magnitude of extreme peaks. The discrepancies between actual and predicted peak values suggest that further improvements in the model are necessary. Enhancements could include incorporating additional data types, refining the model architecture, or employing more sophisticated calibration techniques.

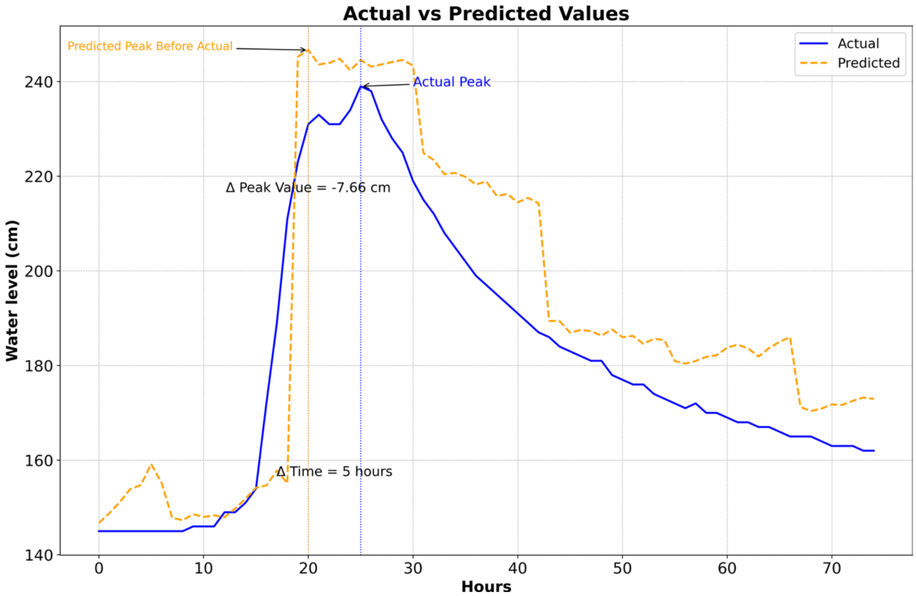

Fig. 8 Peak analysis case 3: actual versus predicted water levels

Additionally, the quality and completeness of the training data significantly affect the model's performance. Inaccurate or missing data can lead to underestimation or overestimation of extreme events, as observed in the cases analyzed. Therefore, ensuring high-quality, comprehensive datasets is crucial for training reliable predictive models.

In summary, while the model shows promise in early peak prediction, which is beneficial for flood management, ongoing efforts to address its limitations are essential to enhance its precision and reliability in predicting extreme hydrological events.

5 Discussion

The deep learning model developed in this study demonstrates significant capabilities in predicting water levels, particularly in forecasting extreme values. The model's performance metrics-such as MSE of 18.07, RMSE of 4.25, MAE of 1.82, MAPE of 1.11%, and R^2 of 0.8785-highlight its robustness and accuracy. These results indicate that the model can reliably predict both regular and extreme water levels, which is crucial for flood management and early warning systems.

While the deep learning model developed in this study focuses on predicting water levels rather than directly forecasting flood events, its outputs are foundational for flood prediction systems. Accurate water level predictions are a strong proxy for assessing flood risks, especially when combined with additional variables like precipitation, soil moisture, and historical flood data. However, it is crucial to acknowledge that not every peak in water level necessarily results in a flood. Regional factors, such as local topography, land use, and historical flood events, must be considered to accurately interpret these predictions in the context of flood risk. Future research should focus on integrating these regional characteristics into the model, enhancing its utility in flood forecasting and management. Additionally, it would be beneficial to compare the model's predictions with past flood events (e.g., those between 2008–2020) to better understand how changes in water levels correlate with actual flood occurrences. This comparison could also involve calculating return periods for precipitation or water levels, offering a more comprehensive approach to flood risk assessment.

The superior performance of our model in predicting extreme water levels is primarily attributed to the strategic integration of LSTM, GRU, Conv1D, and Multi-Head Attention mechanisms. These components were specifically chosen to address the non-linear and complex temporal dependencies characteristic of hydrological data. The LSTM and GRU layers are particularly effective in capturing sequential dependencies, which are crucial for understanding the progression of water levels over time. The Conv1D layers were incorporated to extract local patterns, such as hourly fluctuations, while the Multi-Head Attention mechanism enhances the model's ability to focus on the most relevant features in the data, allowing it to prioritize critical periods leading up to extreme events. This combination of layers ensures that the model not only captures the temporal dynamics but also identifies and emphasizes key features that signal the onset of extreme hydrological conditions (Li et al. 2022; Zhang et al. 2022; Luppichini et al. 2023; Kratzert et al. 2019; Lin et al. 2023).

When comparing the performance of our model with other international studies, it is evident that hybrid deep learning models continue to demonstrate superior accuracy in flood forecasting across different regions. For instance, Zhang's study on the Humber River in Toronto employed a Spatio-Temporal Attention LSTM (STA-LSTM) model, which outperformed traditional LSTM and CNN-LSTM models, particularly in predicting discharge levels 12 h ahead (Zhang et al. 2022). This model's effectiveness in capturing temporal dependencies aligns with our model's successful integration of LSTM and attention-based mechanisms, highlighting the importance of hybrid architectures in improving forecast accuracy under varying conditions. Similarly, the comprehensive review by Luppichini et al. (2023) underscores the critical role of deep learning models, including LSTM and GRU, in enhancing flood prediction accuracy by managing temporal dependencies in river flow data. Furthermore, the FastGRNN-FCN model developed by Dong et al. (2020) for urban flood prediction in Harris County, Texas, demonstrated a high level of precision with an F-measure of 0.792, emphasizing the utility of combining recurrent neural networks with fully convolutional networks for accurate spatial-temporal flood propagation prediction. These international studies corroborate the findings of our research, reinforcing the effectiveness of hybrid deep learning models in diverse hydrological contexts and supporting the argument that such models are crucial for advancing flood prediction and management globally.

Accurate predictions of extreme water levels can significantly aid in mitigation strategies. By providing a 12-hour advance prediction, the model allows for timely alerts and preparedness measures, potentially reducing the impact of flooding. The early detection of peaks, as illustrated in the case analyses, underscores the model's practical value in anticipating critical events and initiating appropriate responses. However, despite these strengths, there are some limitations. While the model excels in predicting extreme water levels, it has a tendency to slightly overestimate these values. This overestimation, although within an acceptable range, suggests that the model is calibrated to prioritize the accurate prediction of high-risk events, which is crucial for early warning systems. Nevertheless, this can be easily adjusted using a simple statistical model to fine-tune the predictions. Additionally, instances of underestimation were observed, but these were predominantly in non-critical scenarios where the actual water level was below dangerous thresholds. This indicates that while the model is highly sensitive to potential floods, there is room for improvement in ensuring that even lower-level water predictions are accurate. Future work could focus on refining the model to balance both overestimation and underestimation, ensuring even greater reliability across all water levels.

When compared to traditional hydrological models such as Auto-Regressive Integrated Moving Average (ARIMA) or basic neural networks, the deep learning model developed in this study shows notable improvements. Traditional models often struggle with the non-linear and complex relationships inherent in hydrological data. In contrast, the combination of LSTM, GRU, Conv1D, and Multi-Head Attention mechanisms in this model effectively captures temporal patterns and dependencies, enhancing prediction accuracy. This model's performance, with an R^2 of 0.8785, represents a substantial improvement compared to existing flood prediction models. For instance, (Nanda et al. 2016)'s ARMAX model reported an R^2 of 0.77, and (Choubin et al. 2018)'s classification and regression trees (CART) model achieved an R^2 of 0.8. Additionally, (Ha et al. 2021) used deep neural networks, including stacked LSTM, ConvLSTM encoder-decoder, and ConvLSTM encoder-decoder GRU, to predict Yangtze River streamflow and achieved R^2 values between 0.84 and 0.88. Their study highlighted the model's ability to predict flood peaks more accurately by integrating El Niño-Southern Oscillation (ENSO) data. These promising results are particularly notable given the absence of any structures along the river to regulate water flow or levels. This success can be attributed to the model's sophisticated architecture and a custom loss function focused on extreme values, ensuring more accurate predictions in critical scenarios.

6 Conclusion

This study successfully developed a deep learning model tailored for accurate water level predictions, particularly for forecasting extreme events. By integrating advanced architectures such as LSTM, GRU, Conv1D, and Multi-Head Attention layers, the model effectively captured the complex temporal dependencies inherent in hydrological data. The model achieved high accuracy, with significant performance metrics that underscore its reliability in predicting water levels, even in river systems lacking regulatory structures.

The innovative combination of these deep learning techniques marks a significant advancement in hydrological forecasting, addressing key challenges in predicting extreme events. This model not only enhances flood management and early warning capabilities but also offers a valuable tool for disaster preparedness in regions with similar hydrological challenges.

While the model showed strong performance, especially in predicting extreme values, future work should aim to refine its accuracy and broaden its applicability. This includes integrating additional data types, such as soil moisture and meteorological forecasts, and refining the model architecture and calibration techniques. Expanding the dataset to encompass diverse hydrological scenarios and testing the model in varied regions will help validate its generalizability and enhance its role in comprehensive flood risk assessment.

In summary, this study contributes valuable insights and practical tools to the field of hydrological forecasting, offering a robust and adaptable solution for anticipating and managing extreme water level events, ultimately supporting more resilient flood management practices.

Author contributions All authors contributed to the study conception and design. Material preparation, data collection, model development, training, and analysis were performed by Wael Almikaeel. The first draft of the manuscript was written by Wael Almikaeel, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availibility The datasets generated during and/or analyzed during the current study are stored in a public repository on GitHub and are available at https://github.com/Wael-Mikaeel/Hydro-informer.

Code availability The code used to support the findings of this study is stored in a public repository on GitHub and is available at https://github.com/Wael-Mikaeel/Hydro-informer.

Declarations

Conflict of interest The authors declare no conflict of interest.

Ethical approval and consent to participate Not applicable.

Consent for publication Not applicable.

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